MYSQL

1. Overview

- ✓ SQL – Structured query language is a special perpose programming language designed for managing data held in a RDBMS (Relational Database management System).
- ✓ SQL like languages can also used in RDBMS or in NoSQL (Not Only SQL)

2. Identifier :

- ✓ Same like variable in other programming languages.
- ✓ Difference is this is only identifier not to store data like variable.
- ✓ Can b start with (a-z) letter , (0-9) digits and underscore (_) and must start with letter.
- ✓ Oracle (convert to UPPER case)
- ✓ Postgre (convert to lower case)

3. Data Types :

To store a specific types of value in a identifier (column) we use Data Type.

- ➢ Numeric :
  - DECIMAL(precision)
  - Numeric(precision)
- ➢ Float
- ➢ Integer [smallint (2-byte), tinyint (1 –byte) ,int(4-byte),bigint(8-byte)]
- ➢ Binary
- ➢ Char(n)
- ➢ Varchar(n)/varchar2
- ➢ NCHAR(n)/ NVARCHAR(n)
- ➢ Date
- ➢ Timestamp

4. Schema :

Schema is a collection of tables with rows and columns and a separate query can be written for the **schemas** like databases. Actually, **schema** meant a template in **MySQL**. they define size, type, a grouping of information. The **schemas** have database objects like views, tables, and privileges

Need :

- ✓ To store type of info.
- ✓ Same name tables can be present in different schema.

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5. Create table :

In RDBMS we always store information in table.

- Create a new table :

Syntax :

Create table <table_name>

(indentifier_1 dataType(precision) <constraints> , …)

Exm :

```
CREATE TABLE Employees(
    Id int identity(1,1) primary key not null,
    FName varchar(20) not null,
    LName varchar(20) not null,
    PhoneNumber varchar(10) not null
);
```

```
CREATE TABLE Cities(
    CityID INT IDENTITY(1,1) NOT NULL,
    Name VARCHAR(20) NOT NULL,
    Zip VARCHAR(10) NOT NULL
);
```

- Create table with Foreign Key :

Syntax : need to add in query

```
CityID INT FOREIGN KEY REFERENCES Cities(CityID)
```

Example :

Parent Table :

```
CREATE TABLE Cities(
    CityID INT IDENTITY(1,1) NOT NULL,
    Name VARCHAR(20) NOT NULL,
    Zip VARCHAR(10) NOT NULL
);
```

Child table :

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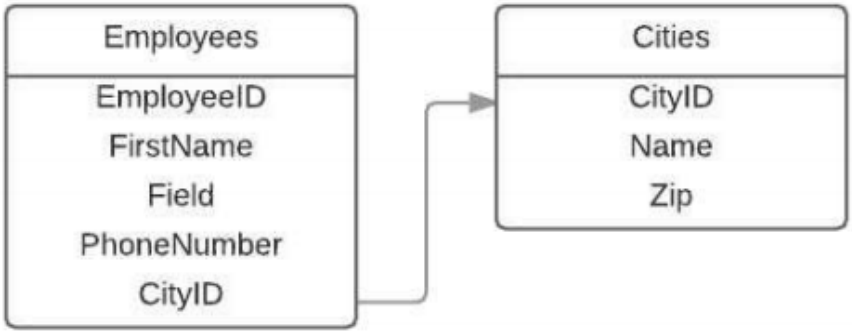
```sql
CREATE TABLE Employees(
    EmployeeID INT IDENTITY (1,1) NOT NULL,
    FirstName VARCHAR(20) NOT NULL,
    LastName VARCHAR(20) NOT NULL,
    PhoneNumber VARCHAR(10) NOT NULL,
    CityID INT FOREIGN KEY REFERENCES Cities(CityID)
);
```

Concept :



- Create table from an Other table :

Create table new_table_name as select * from old_table_name;

6. Insert

When we want to insert Data in to created table . we use insert statement

Insert new row :

Syntax:

Insert into <table> (column_name) values (values separated by ‘,’)

Exm :

Insert into employees (employeeId ,FirstName ,field , PhoneNumber ,cityID) values (100,’sagar’,’RD’,’12345’,91)

- Insert only specific values :

Insert into employees (employeeId ,FirstName ,field ,cityID) values (100,’sagar’,’RD’ ,91)

We can insert multiple value at a time .

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- Insert data from one table to an other :

Insert into tgt_table_name select * from src_tbl_name;

7. update

- Update specific value :

Syntax :

Update <table_name> set <column_name>=<value> where <condition>

Exm :

```
UPDATE Cars
SET TotalCost = TotalCost + 100
WHERE Id = 3 or Id = 4
```

- Update all rows : (use carefully)

```
UPDATE Cars
SET Status = 'READY'
```

8 . NULL :

Null meaning is nothing.in simple way “the absence of value”.null and ‘null’ or ‘’ or 0 is not same.

- Filtering recording based on null:

  Syntax :

  Select * from table_name where column_name is null;

- Nullable column in table :

  When creating table it is possible to declare column can be nullable or not.

  Exm –

```
CREATE TABLE MyTable
(
    MyCol1 INT NOT NULL, -- non-nullable
    MyCol2 INT NULL      -- nullable
) ;
```

By default every column except PK is Nullable .

When we try to assign null value to not nullable column will lead to error.

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```sql
INSERT INTO MyTable (MyCol1, MyCol2) VALUES (1, NULL) ;  -- works fine

INSERT INTO MyTable (MyCol1, MyCol2) VALUES (NULL, 2) ;
        -- cannot insert
        -- the value NULL into column 'MyCol1', table 'MyTable';
        -- column does not allow nulls. INSERT fails.
```

- Updating fields to Null :

  Updating the value as null is same as updating other values .

```sql
UPDATE Employees
SET ManagerId = NULL
WHERE Id = 4
```

- Inserting fields with Null values :

  If value of any attribute is not present you can pass there null

```sql
INSERT INTO Employees
    (Id, FName, LName, PhoneNumber, ManagerId, DepartmentId, Salary, HireDate)
VALUES
    (5, 'Jane', 'Doe', NULL, NULL, 2, 800, '2016-07-22') ;
```

9.Drop :

By using drop we can drop schema or table .

Syntax :

Drop table/database <database_name/table_name>

Exm :

Drop table student;

Drop database mydb;

10.Truncate :

If we want to keep table structure as it is but want to delete all data that time we use truncate.

Syntax :

Truncate table <table_name>

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11.Comments :

Ignore the line from script .

- Single line comment :

“ - - “

- MultiLine comment :

/* msg */

Run script :

```
source C:/bigdata/Database/test.sql
```

Delete row from table :

When we want to delete some records from table that time we use delete .

Syntax :

Delete from student where srno=3;

12. SELECT : (vvvvv Imp)

- Select particular column :

  Select name from student ;

- Select all columns :

  Select * from student ;

- Dot (.) notation :

  When we want to select data from an other database .

Select * from mydb.emp;

- Column alias :

  Select sal * 12 as annual_sal from table emp;

- Select individual columns :

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Select col1,col2,col3 from emp;

- Select specific number of records :

  Select * from student limit 2 ;

- Select with order :

  Select * from employee order by name asc/desc;

- Select with case :

  Select case when mark > 70 then ‘pass’ else ‘fail’ end as status from student;

- Select with condition :

  Select * from emp where city=’pune’

- Select with multiple condition :

  Select * from emp where city=’pune’ and did=’10’;

- Select with multiple conditions :

  Select * from emp where city in (‘pune’,’mumbai’);

- Count the number of records :

  Select count(1) from student:

- Selecting with null :

  Select * from emp where dept is null;

- Select distinct values :

  Select distinct * from emp;

  Select distinct city from emp;

- Select with join:

- Select with aggregate function:

- Select with group by :

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13.Function :

- String Function :
- Concatenations : (A||'_'||B)

```
SELECT 'Hello' || 'World' || '!'; --returns HelloWorld!
```

- Length : length(name)
- Trim : trim empty spaces :
  - Ltrim(‘ Hello ’)
  - Rtrim(‘ Hello ’)
  - trim(‘ Hello ’)
- upper
- lower
- split :

```
SELECT value FROM STRING_SPLIT('Lorem ipsum dolor sit amet.', ' ');
```

Result:

```
value
-----
Lorem
ipsum
dolor
sit
```

- Replace :

Syntax:

REPLACE(String to search , String to search for and replace , String to place into the original string)

Example:

```
SELECT REPLACE( 'Peter Steve Tom', 'Steve', 'Billy' ) --Return Values: Peter Billy Tom
```

- Substring :

Syntax is: SUBSTRING (string_expression, start, length). Note that SQL strings are 1-indexed.

```
SELECT SUBSTRING('Hello', 1, 2) --returns 'He'
SELECT SUBSTRING('Hello', 3, 3) --returns 'llo'
```

- Reverse :

Syntax is: REVERSE (string-expression)

```
SELECT REVERSE('Hello') --returns olleH
```

- Replicate :

Syntax is: REPLICATE (string-expression , integer)

```
SELECT REPLICATE ('Hello',4) --returns 'HelloHelloHelloHello'
```

- **Aggregate Function :**
  - Sum

Select sum(salary) from emp;

```
select sum(salary) TotalSalary
from employees;
```

TotalSalary
2500

- Avg :

```
select city_name, AVG(population) avg_population
from city_population
where city_name = 'NEW YORK CITY';
```

- Count() :

```
SELECT count(*) TotalRows
FROM employees;
```

- Min() :

```
select min(age) from employee;
```

- Max():

```
SELECT MAX(column_name) FROM table_name;
```

- Scalar / Single Row Function :

Now();

Upper /lower etc

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- Group By :

Group by always use with Group function such as count , avg ,sum ,min , max etc.

```
SELECT EmpID, SUM (MonthlySalary)
FROM Employee
GROUP BY EmpID
```

Filter result from group by :

```
SELECT
  a.Id,
  a.Name,
  COUNT(*) BooksWritten
FROM BooksAuthors ba
  INNER JOIN Authors a ON a.id = ba.authorid
GROUP BY
  a.Id,
  a.Name
HAVING COUNT(*) > 1    -- equals to HAVING BooksWritten > 1
```

14.Operator :

- AND OR

Select * from emp where city='pune' and city='Nagar';

Select * from emp where city='pune' or city='Nagar';

- Like (_,%)

Select * from emp where city like 'p%';

- Between … and

- Clause :
- In , not in

Select * from emp where city in ('pune','a.bad')

- Filter result using where and having :

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Describer : to see the table details .

Syntax :

Desc table_name or Describe table_name

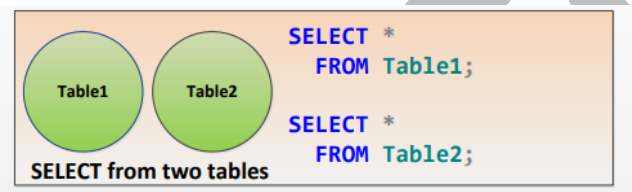
15.Join

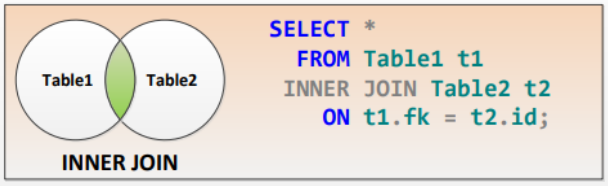
When we want to select data from more than one table . we use join

- Types of join :
  - Inner join
  - Left outer
  - Right Outer
  - Full outer
  - Self join
  - Cross join

Inner join :

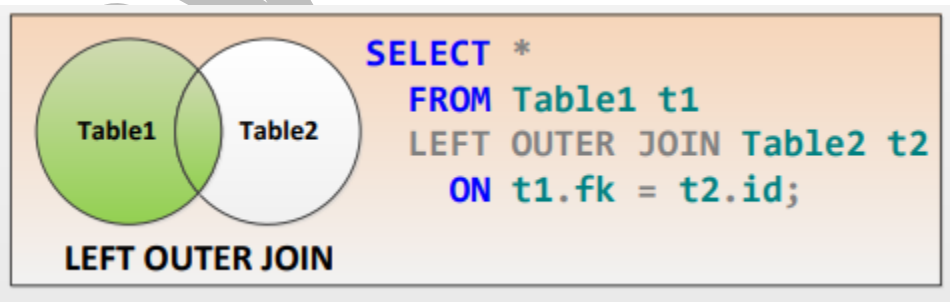
It shows only matching records





Left Outer join :

It gives all records from left table and all matching records.



Right Outer Join :

It gives all records from right table and all matching records.

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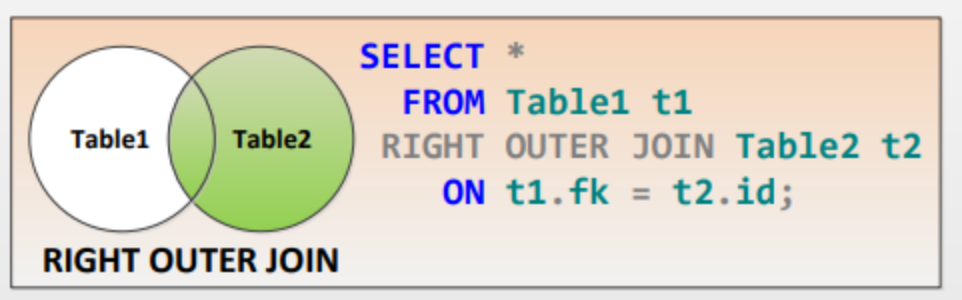
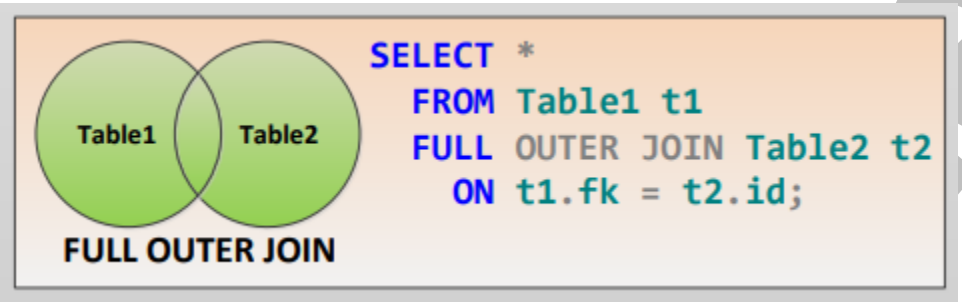


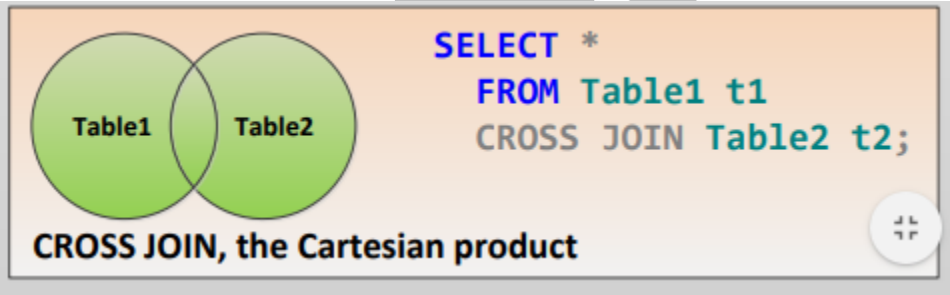
Full Outer join :

Its gives all record form both tables with matching records.



Cross join :

It gives the result as cross product.



Self join :

When we want to join table on itself we use self join.

16. Set Operations:

Union

UNION is used to combine the results of two or more `SELECT` statements. However **it will eliminate duplicate rows from its resultset.** In case of union, number of columns and datatype must be same in both the tables, on which UNION operation is being applied.

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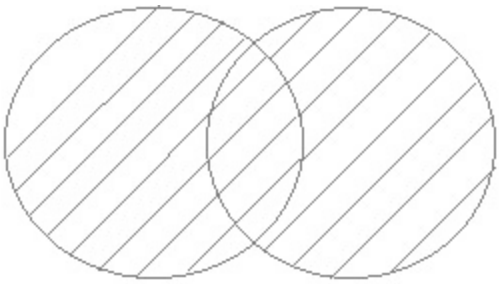
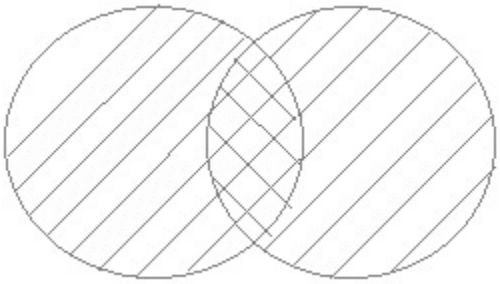


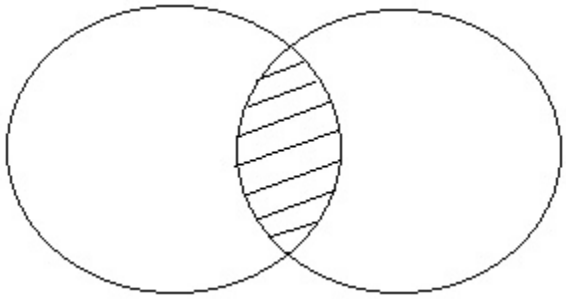
UNION ALL :

This operation is similar to Union. But it also shows the duplicate rows.



INTERSECT

MySQL does not support INTERSECT operator.



MINUS

The Minus operation combines results of two `SELECT` statements and return only those in the final result, which belongs to the first set of the result.

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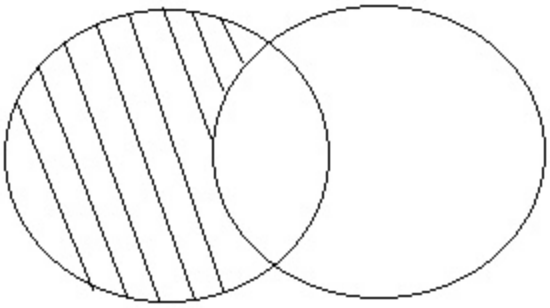
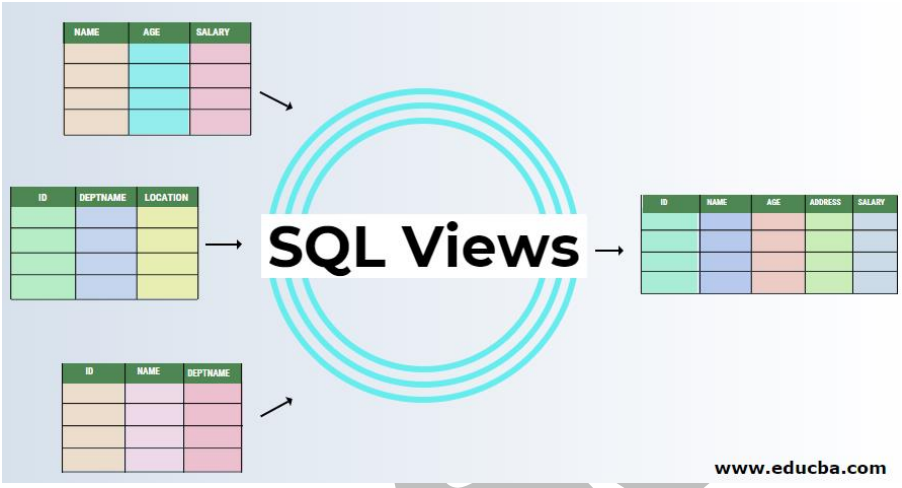


17.View:



We can filter some columns or some rows from the base table.

It is logical template of table.

Not physically stored.

We can run select statement on view like table.

Syntax :

Create view view_name as <select_statement>

- Simple view (1 million row)
  Filter only from the base table.
  Create view v1 as select eid,city from emp where city='pune';
- Complex View :
  If a view have complex query (join,aggregation and subquery)
  Create view v2 as select a.id as eid ,b.dname as rname form emp a join dept b on a.did=b.did

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- Materialized Views:

A view whose results are physically stored and must be periodically refreshed in order to remain current. Therefore useful for storing the result of complex, long running queries when realtime result are not required.

It will work in Oracle and Postgre.

Create materialized view v2 as select * from emp where city='pune'.

18 . Alter :

Add column :

```
ALTER TABLE Employees
ADD StartingDate date NOT NULL DEFAULT GetDate(),
    DateOfBirth date NULL
```

Drop column:

```
ALTER TABLE Employees
DROP COLUMN salary;
```

Add primary key :

```
ALTER TABLE EMPLOYEES ADD pk_EmployeeID PRIMARY KEY (ID)
```

Drop Constraint :

```
ALTER TABLE Employees
DROP CONSTRAINT DefaultSalary
```

Alter Column :

```
ALTER TABLE Employees
ALTER COLUMN StartingDate DATETIME NOT NULL DEFAULT (GETDATE())
```

19. Subqueries :

Query with in query is nothing but subquery.It acts similarly to a temporary table that is generated during the execution of query and lost afterword .

```
SELECT *
FROM Employees
WHERE Salary = (SELECT MAX(Salary) FROM Employees)
```

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